EXPERIMENT 8

Nmap Service Version & OS Detection Commands (-sV, -A, -O)

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AIM:

To execute and demonstrate the functionality, syntax, and outputs of Nmap service version detection (-sV), aggressive scan (-A), and operating system detection (-O) commands on a target host in Kali Linux Operating System.

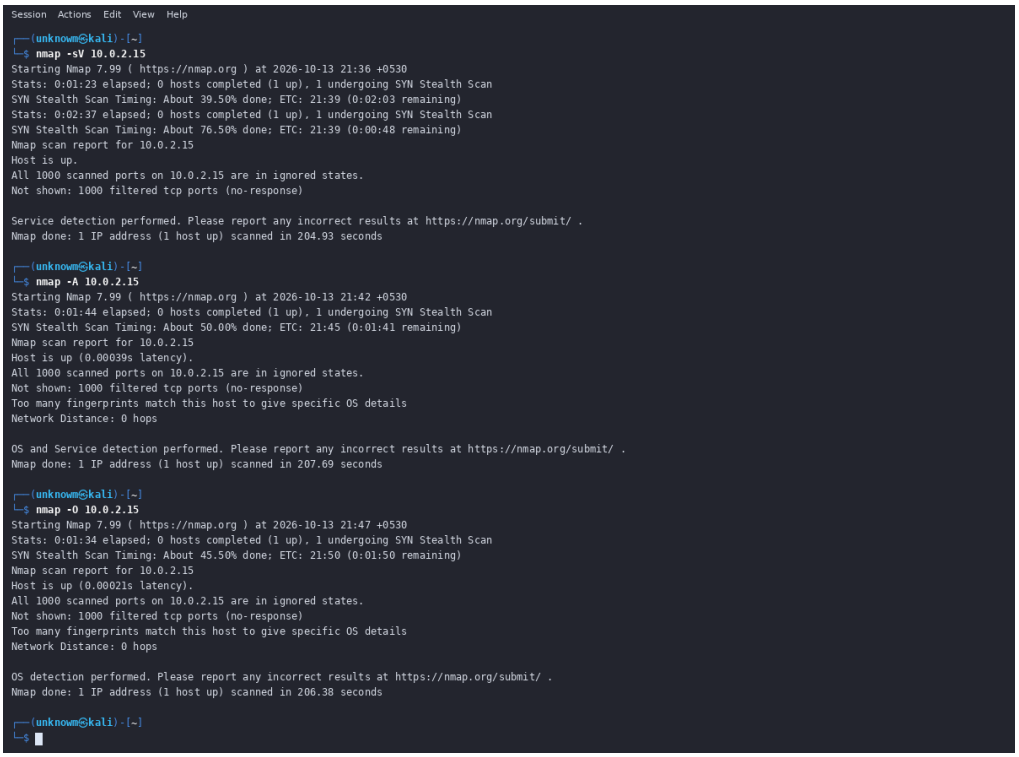
COMMANDS AND DESCRIPTION:

1. nmap -sV <target> (Service Version Detection):
• *Description:* Probes open ports to determine service and version info running on each port using Nmap's service fingerprints database.
• *Syntax:* nmap -sV [options] <target_IP>
• *Options:* --version-intensity <0-9> (probe intensity), --version-light (faster light scan).

2. nmap -A <target> (Aggressive Scan):
• *Description:* Enables aggressive scan mode, which combines OS detection, service version scanning, script scanning (traceroute), and traceroute.
• *Syntax:* nmap -A [options] <target_IP>

3. nmap -O <target> (Operating System Detection):
• *Description:* Uses TCP/IP stack fingerprinting by sending specific probe packets and comparing responses against Nmap's OS database.
• *Syntax:* nmap -O [options] <target_IP>
• *Options:* --osscan-guess (guess OS aggressively when no exact match is found).

OUTPUT:



RESULT:

Thus, the Nmap commands for service version detection (-sV), aggressive scanning (-A), and OS detection (-O) were successfully executed on the target host in Kali Linux, and the scan results were analyzed and verified.